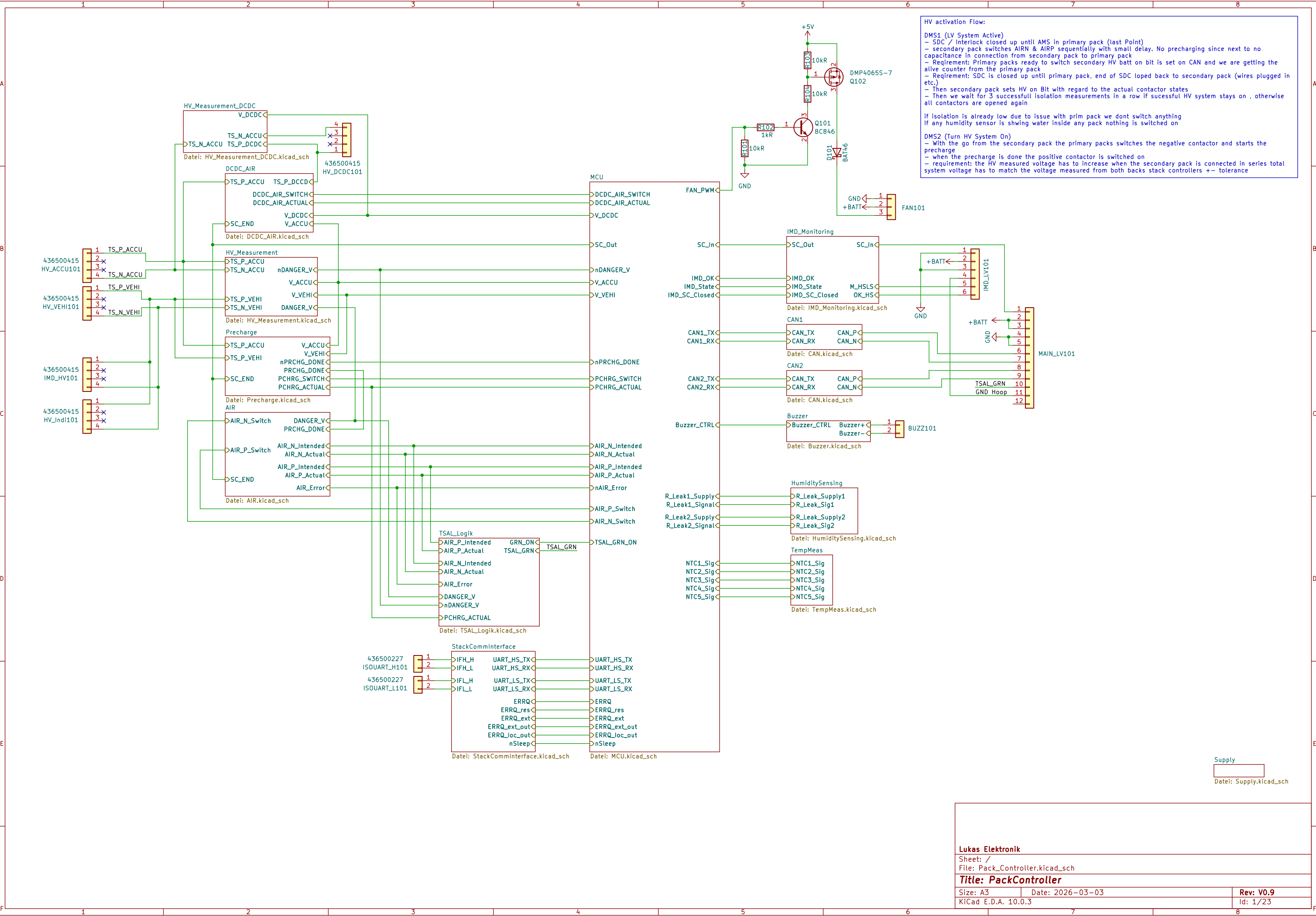
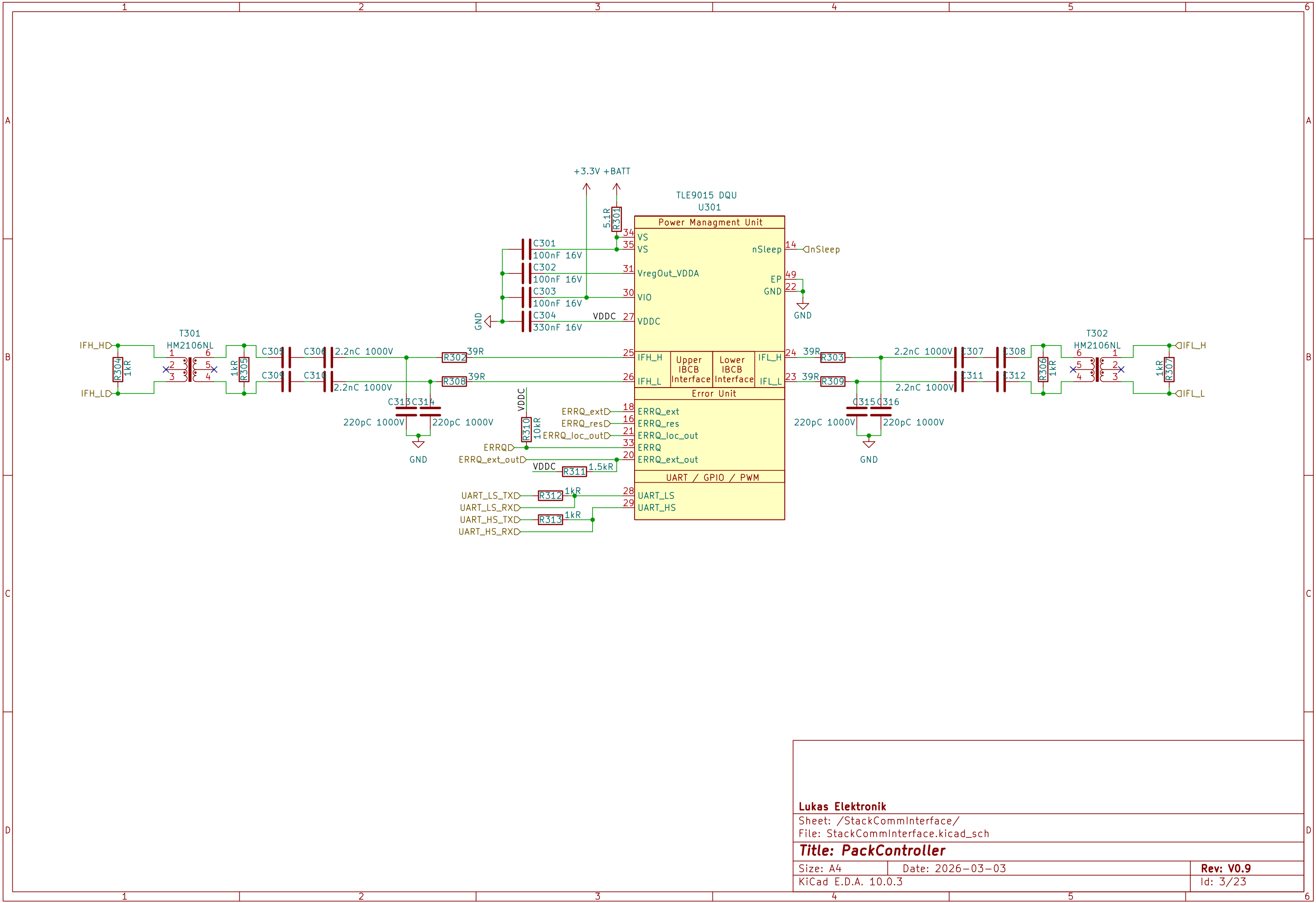
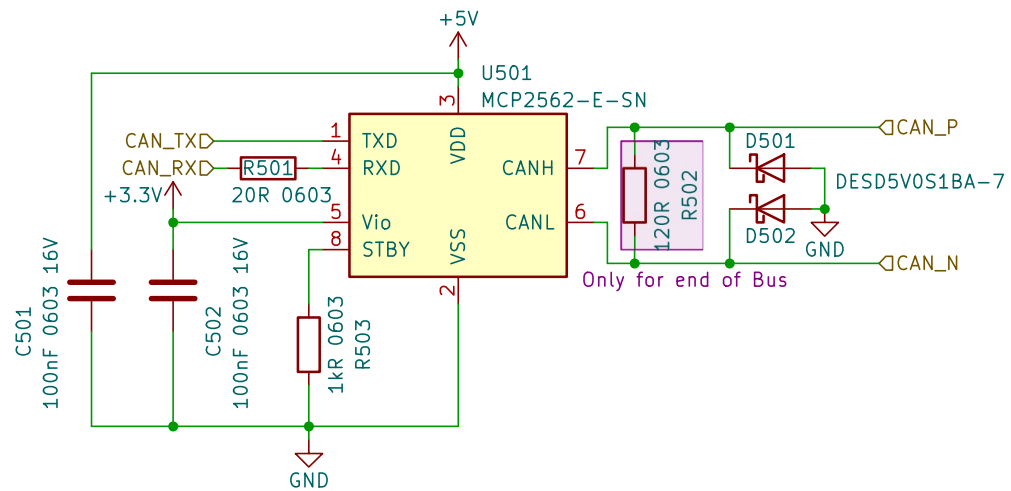








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Sheet: /CAN1/
File: CAN.kicad_sch

Title: PackController

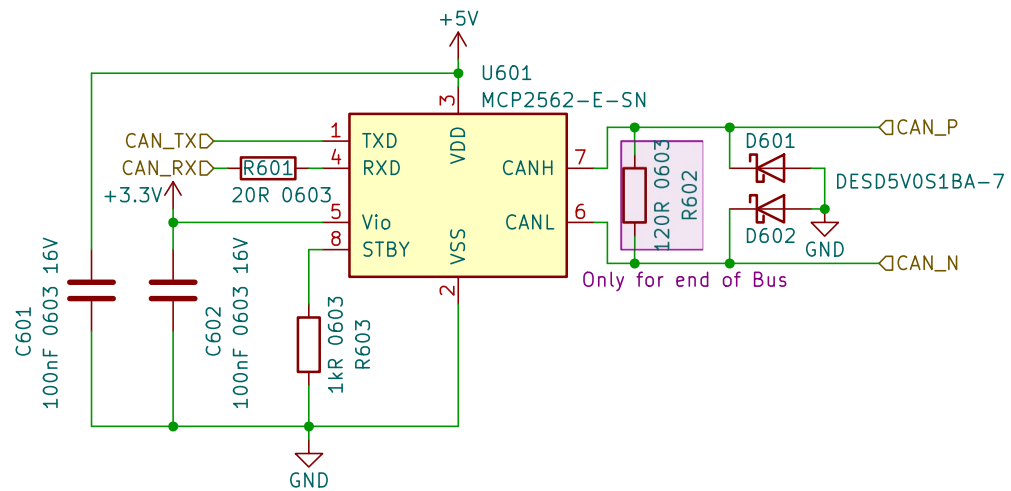
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 5/23



Lukas Elektronik

Sheet: /CAN2/
File: CAN.kicad_sch

Title: PackController

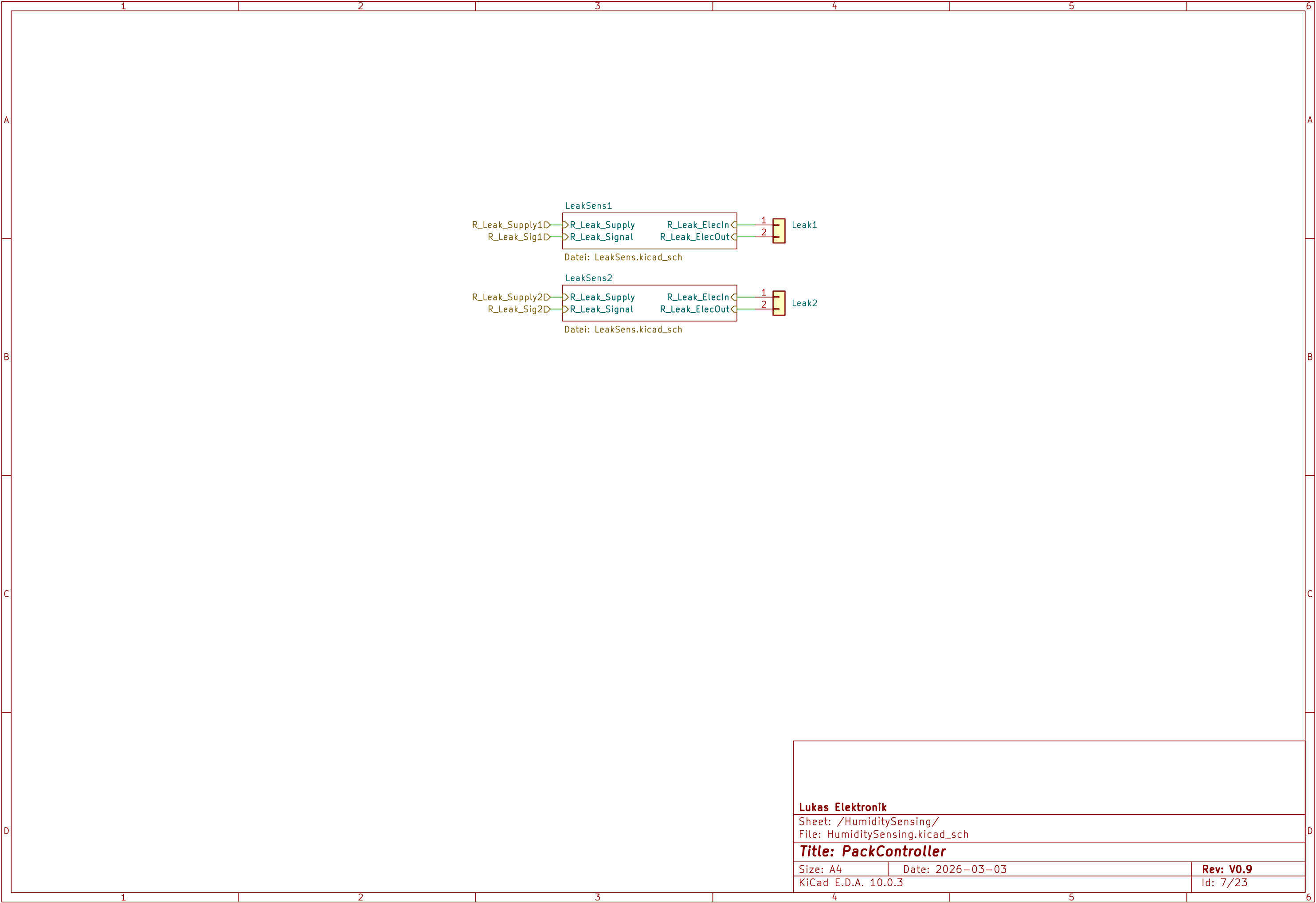
Size: A4

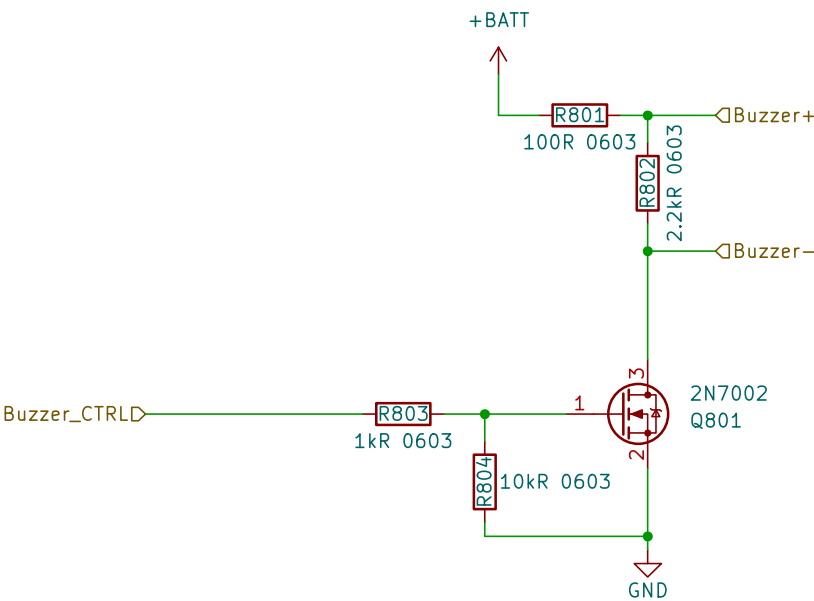
Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 6/23





Passive piezo buzzer driver.
Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:
f_PWM ≈ 4 kHz
duty ≈ 50 %

Q15 is a low-side NMOS switch.
R37 discharges the piezo during MOSFET off-time:
 $\tau = R37 * C_{PIEZO} \approx 2.2k * 24nF \approx 53\mu s$.

R36 limits peak current and EMI.
No additional series coupling capacitor required.

Lukas Elektronik

Sheet: /Buzzer/
File: Buzzer.kicad_sch

Title: PackController

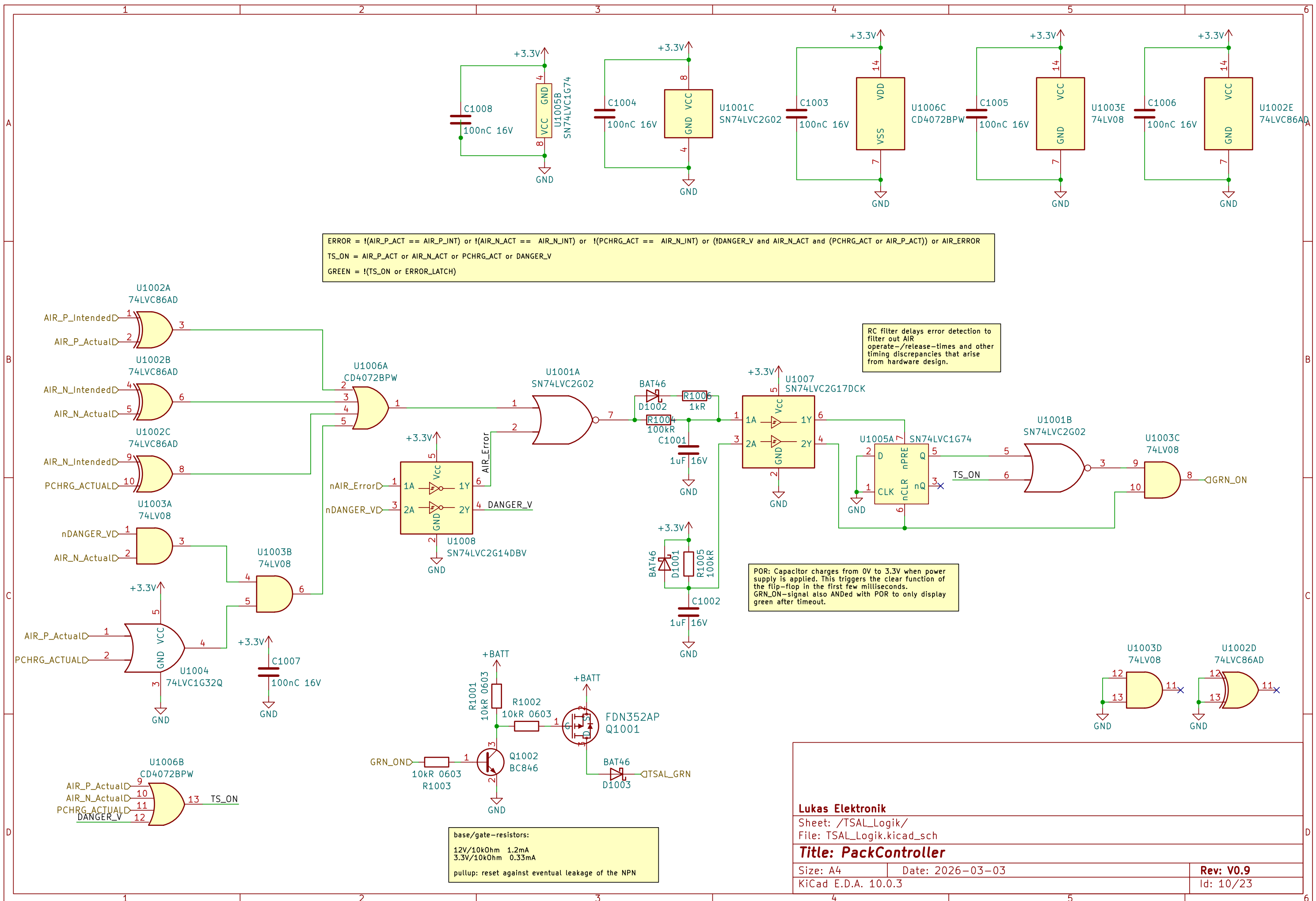
Size: A4

Date: 2026-03-03

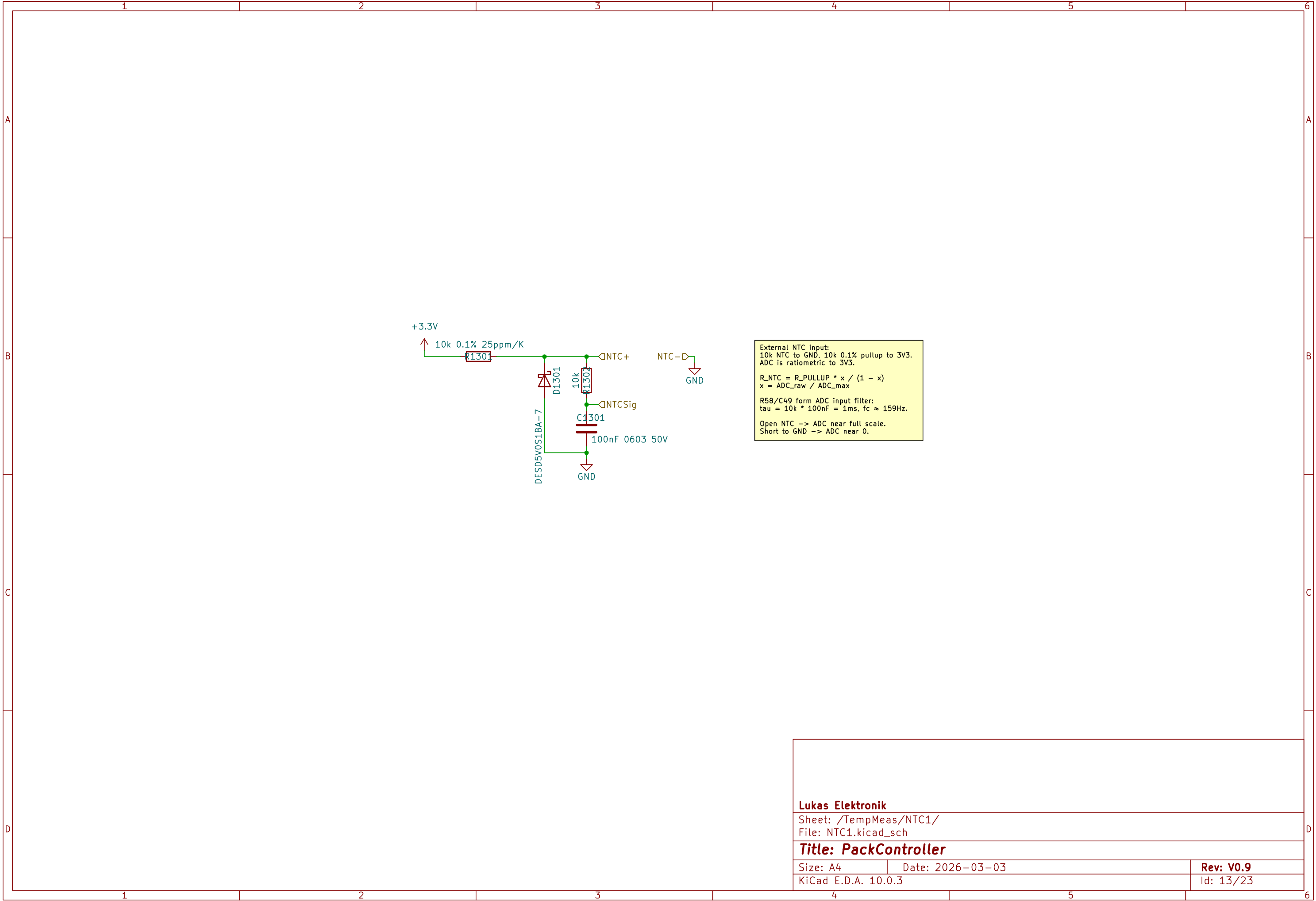
Rev: V0.9

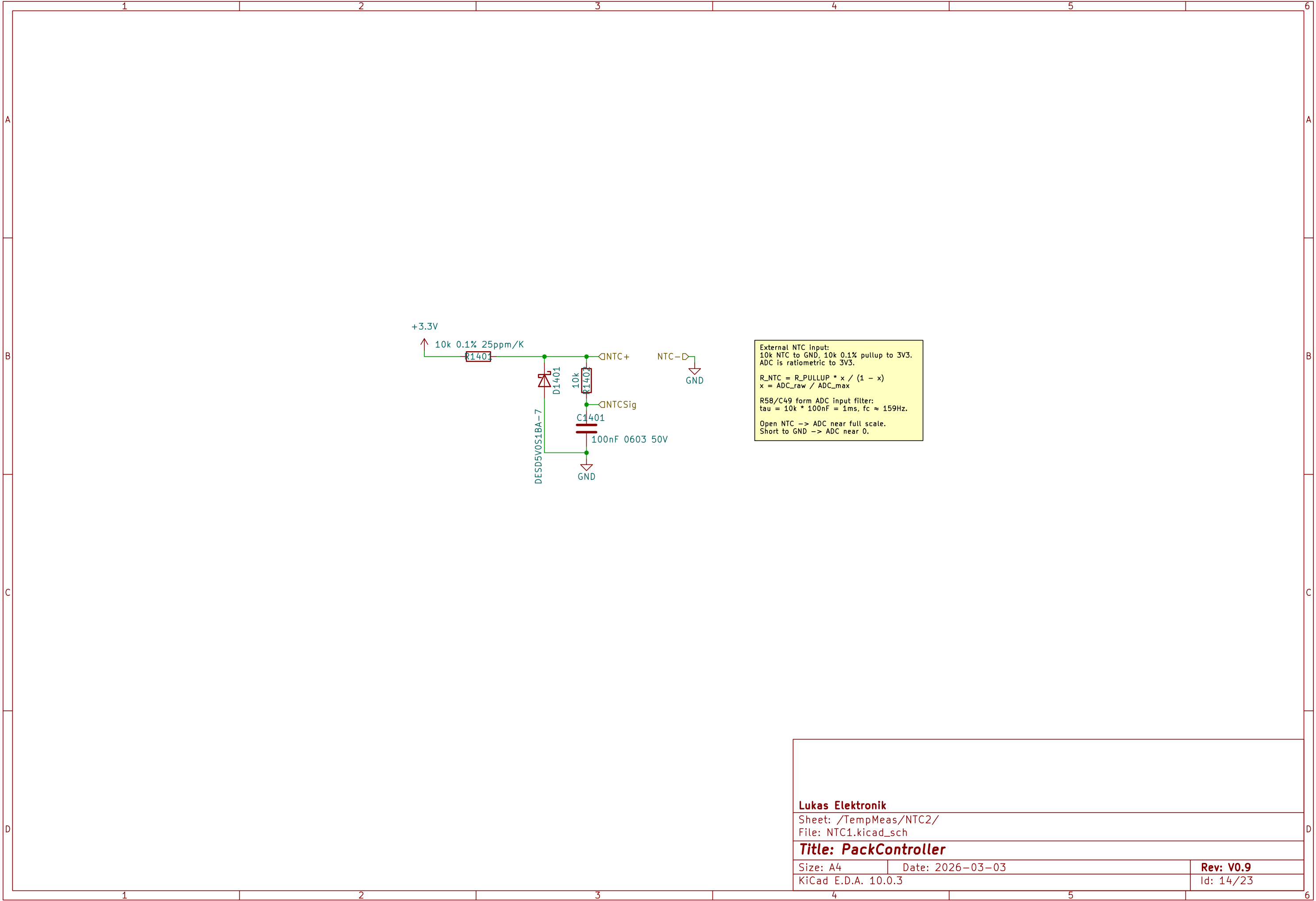
KiCad E.D.A. 10.0.3

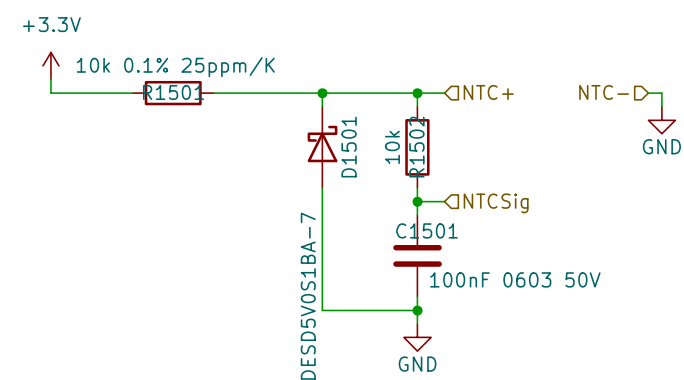
Id: 8/23











External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$
$$x = ADC_{raw} / ADC_{max}$$

R58/C49 form ADC input filter:
 $\tau = 10k \cdot 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC3/
File: NTC1.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

Size: A1	1
KiCad E.D.A. 10.0.3	

Id: 15/23

Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor, parallel to the electrodes, for open-wire detection.

$R_SENSOR = R_LEAK \parallel R_EOL$

Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi-Z. Suggested scan rate: 1Hz.

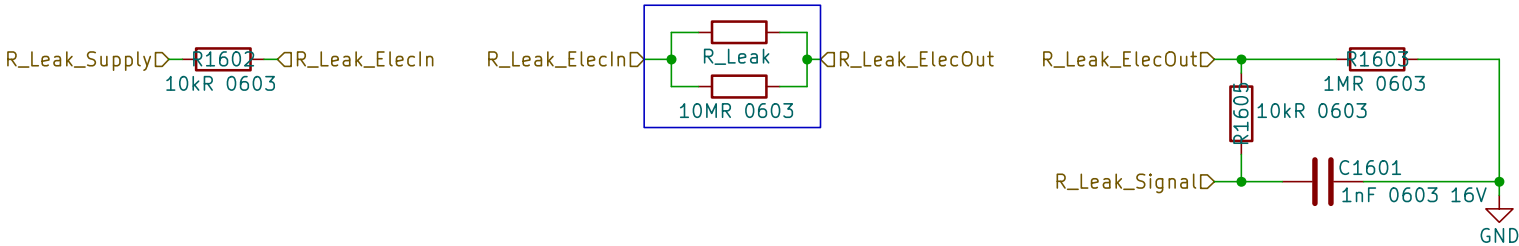
ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$

$R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY - R_ADC_SER$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: $ADC \approx 0$
dry connected: $R_SENSOR \approx 10M$
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: $R_SENSOR < 3M$
leak: $R_SENSOR < 1M$
severe leak: $R_SENSOR < 300k$



Lukas Elektronik

Sheet: /HumiditySensing/LeakSens1/
File: LeakSens.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 16/23

Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor, parallel to the electrodes, for open-wire detection.

$R_SENSOR = R_LEAK \parallel R_EOL$

Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi-Z. Suggested scan rate: 1Hz.

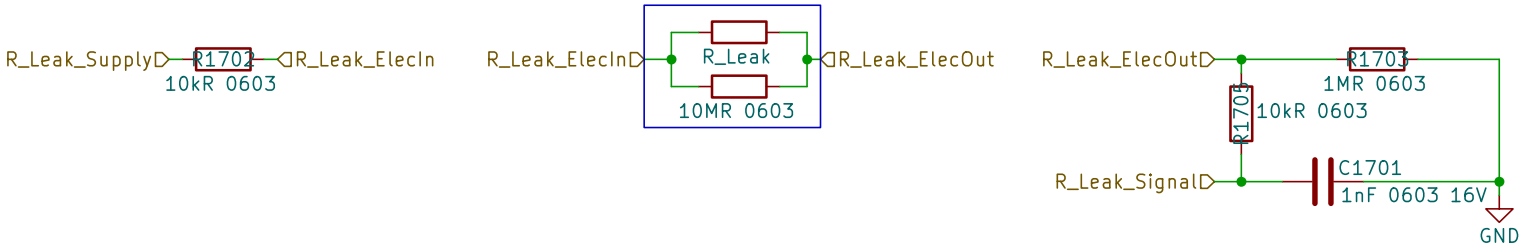
ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$

$R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY - R_ADC_SER$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC $\approx$ 0
dry connected: R_SENSOR $\approx$ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k



Lukas Elektronik

Sheet: /HumiditySensing/LeakSens2/
File: LeakSens.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

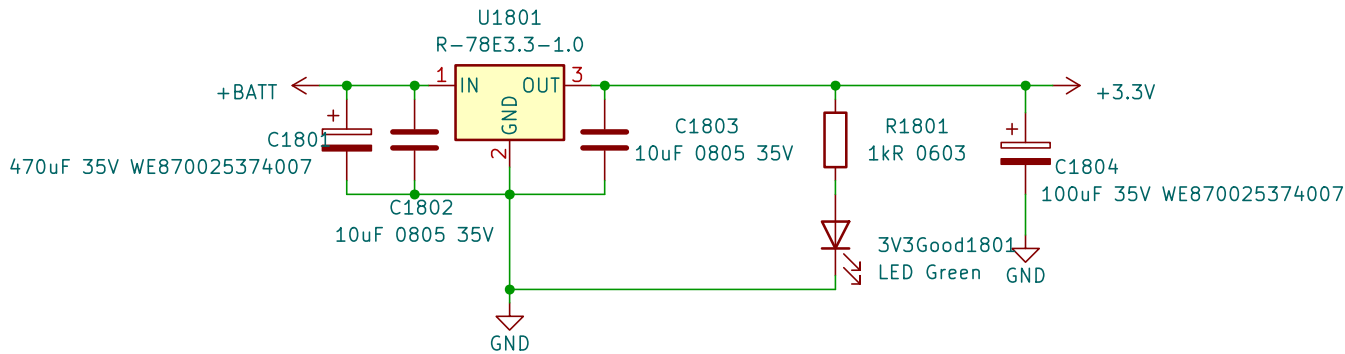
Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 17/23

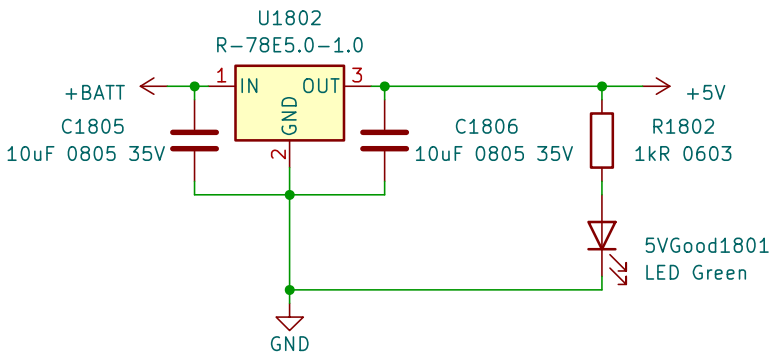
+3V3 Supply worst-case estimate:

Main loads:	
STM32G483 active incl. peripherals:	-50 mA
ADuM3190 VDD1, 2x:	-4 mA
24LC256 EEPROM write:	-3 mA
MCP2562 VIO, 2x worst-case:	-1 mA
ADCM393 comparators + logic:	<1 mA
Pullups, dividers, GPIO drivers, LEDs:	-20...30 mA
Estimated worst-case load:	-80...90 mA
Selected supply:	RECOM 3.3 V / 1 A DC/DC
Design margin:	
1 A / 90 mA $\approx$ 11x	



+5V Supply worst-case estimate:

Main loads:	
TEA1-0505HI isolated DC/DC no-load, 5x:	-120 mA
TEA1 loads incl. ADuM3190 isolated sides:	-25...30 mA
MCP2562 CAN transceivers, 2x dominant:	-140 mA
Misc. reserve / startup / tolerance:	-50 mA
Estimated worst-case load:	-330...400 mA
Selected supply:	RECOM 5 V / 1 A DC/DC
Design margin:	
1 A / 0.4A $\approx$ 2.5x	



Lukas Elektronik

Sheet: /Supply/

File: Supply.kicad_sch

Title: PackController

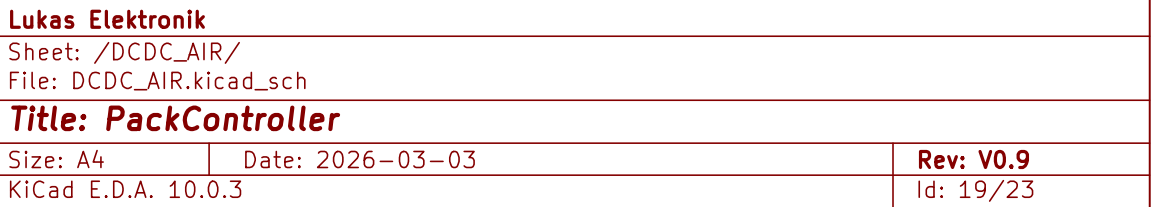
Size: A4

Date: 2026-03-03

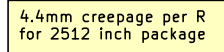
Rev: V0.9

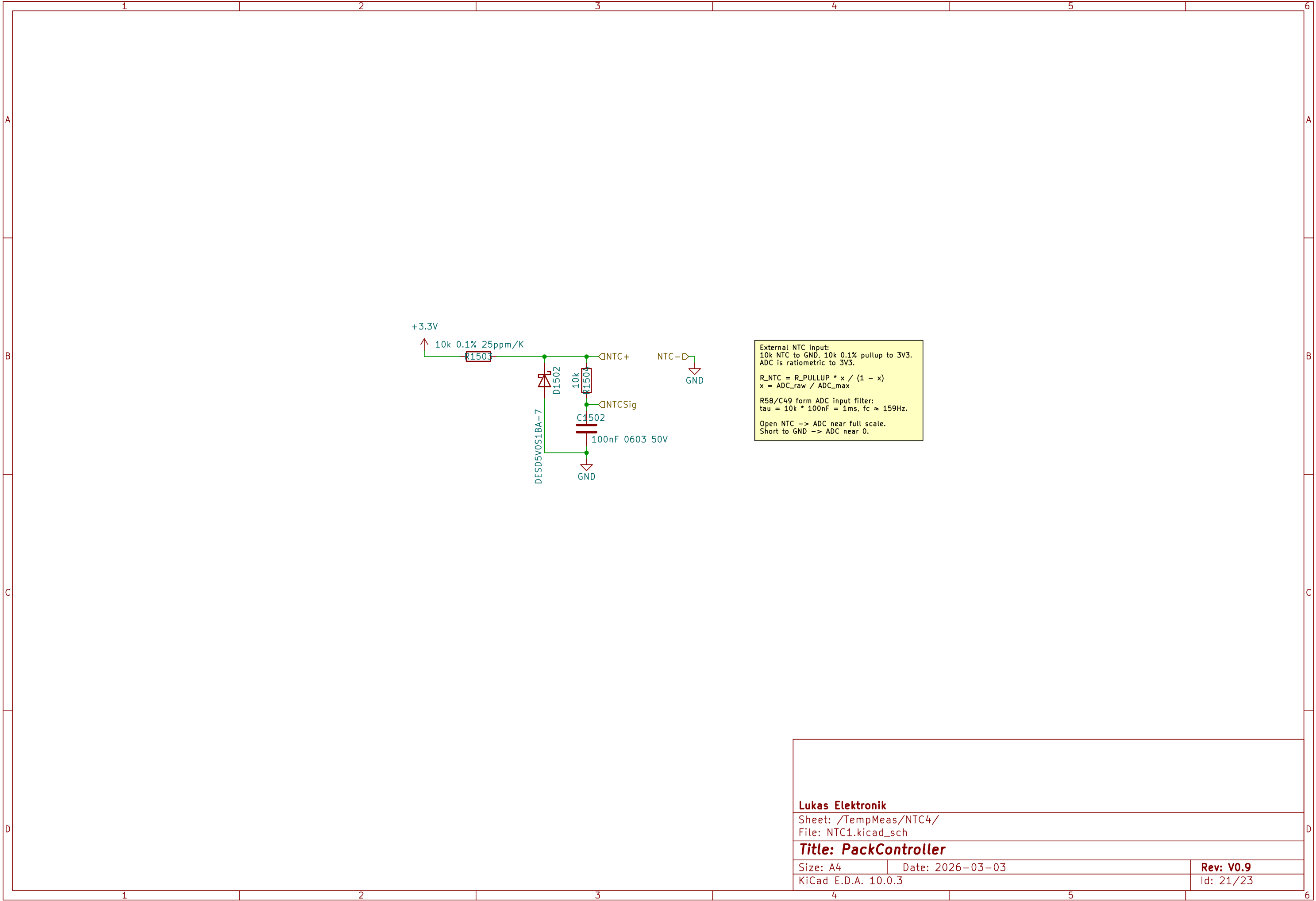
KiCad E.D.A. 10.0.3

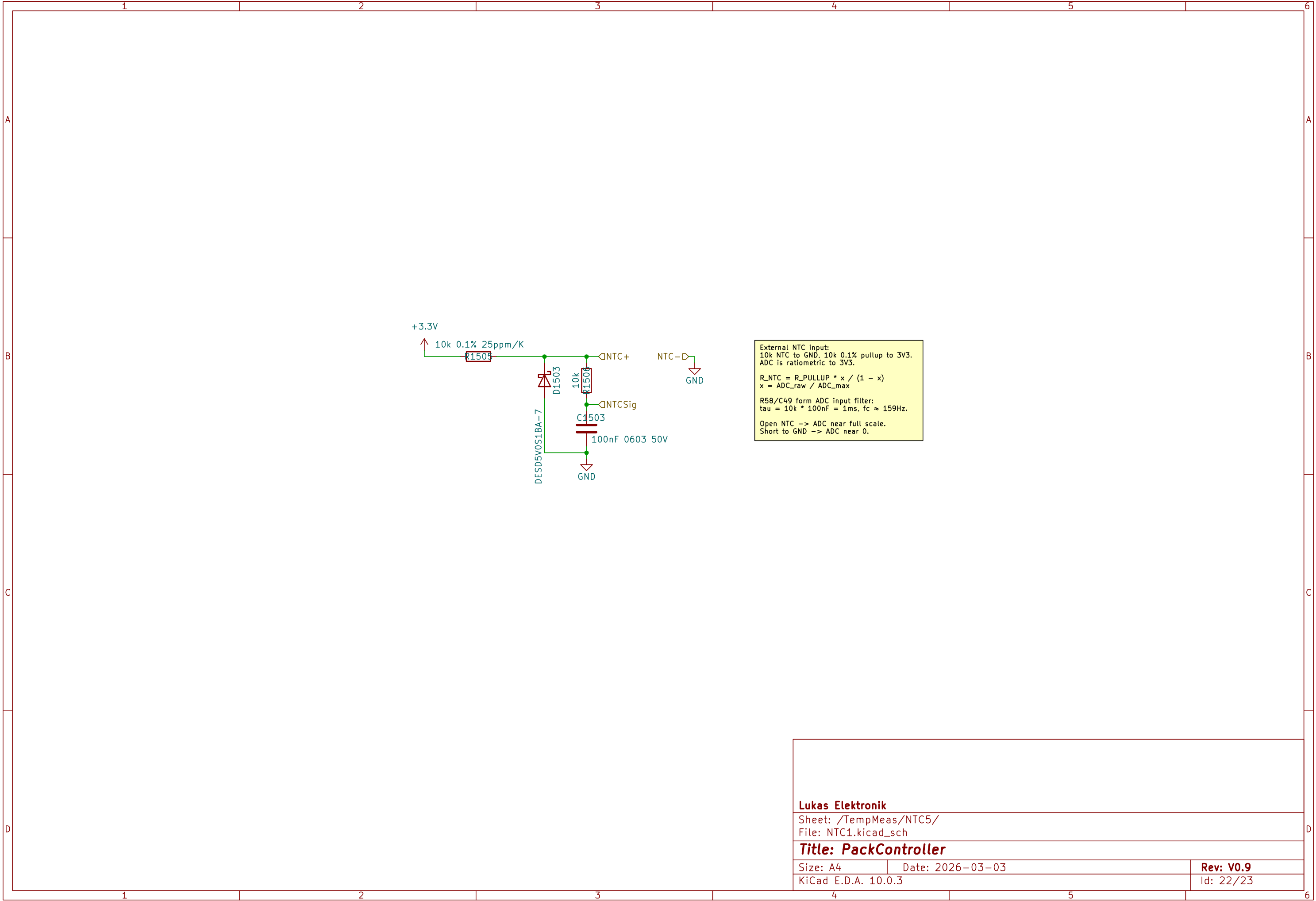
Id: 18/23

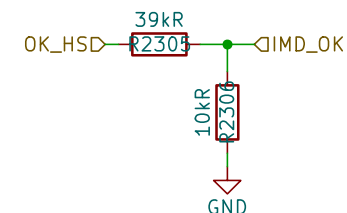
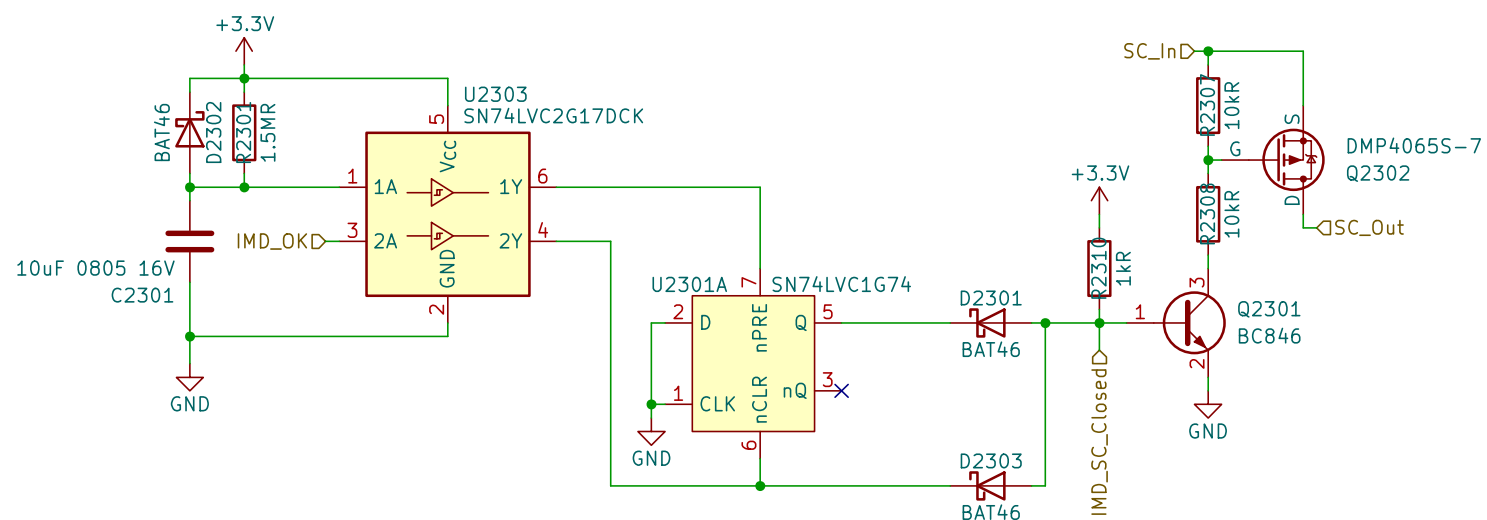


AMC0311R clipping input = 2.56 V
=> HV clipping approx. 924.6 V









During IMD startup, nPRE and nCLR may overlap.
SC_Out is forced disabled by IMD_OK low clamp.
Latch becomes effective only after IMD_OK release.